

NON-INTERVENTIONAL STUDY PROTOCOL

STUDY DETAILS

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TITLE	Ummet Needs and Rates of Key Safety Events Relating to Treatment Use among Newly Treated and Relapsed/Refractory Patients with Multiple Myeloma
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TABLE OF CONTENTS

STUDY DETAILS

TITLE PAGE

LIST OF ABBREVIATIONS

1. SYNOPSIS

2. AMENDMENTS AND UPDATES

3. MILESTONES

4. BACKGROUND AND RATIONALE

4.1. Background

4.2. Rationale

5. RESEARCH QUESTION AND OBJECTIVE(S)

5.1. Research Question(s)

5.2. Objectives

5.2.1. Primary Objective(s)

5.2.2. Secondary objective(s)

5.2.3. Exploratory objective(s)

6. RESEARCH METHODS

6.1. Study Design

6.2. Study Population and Setting

6.2.1. Eligibility Criteria

6.2.1.1. Inclusion Criteria

6.2.1.2. Exclusion Criteria

6.2.2. SOC Categorization

6.2.3. Stratification by patient subgroups

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6.3. Variables

- 6.3.1..... Exposure definitions
- 6.3.2..... Outcome definitions

6.3.2.1.Primary outcomes: key safety events and healthcare

utilisation

6.3.2.2.Secondary objective: treatment patterns and

treatment-related outcomes

6.3.2.3.Exploratory objectives: outcomes, definitions, and

timing

6.3.3..... Confounders and effect modifiers

6.4. Data sources

6.5. Study size

6.6. Data management

6.7. Data analysis

6.7.1..... Primary analyses

6.7.2..... Secondary analyses

6.7.3..... Exploratory/sensitivity analyses

6.7.4..... Data handling conventions

6.8. Quality control and Quality Assurance

6.9. Limitations of the research methods

7. PROTECTION OF HUMAN SUBJECTS

7.1. Ethical approval and subject consent

7.2. Subject confidentiality

8. LEGAL BASIS FOR PROCESSING INDIVIDUAL HUMAN DATA

9. MANAGEMENT AND REPORTING OF ADVERSE

EVENTS/ADVERSE REACTIONS

10.PLANS FOR DISSEMINATING AND COMMUNICATING STUDY RESULTS

11..... REFERENCES

ANNEX 1 LIST OF STAND-ALONE DOCUMENTS

ANNEX 2 CATEGORIZATION OF SOC REGIMENS

ANNEX 3 TABLES

ANNEX 4 FIGURES

ANNEX 5 CODELISTS

ANNEX 6 LOT ALGORITHM

LIST OF ABBREVIATIONS

Abbreviation	Definition
1L	First-Line Regimen
2L	Second-Line Regimen
3L	Third-Line Regimen
4L	Fourth-Line Regimen
ADC	Antibody-Drug Conjugate
AE	Adverse Event
AST	Aminotransferase
BCMA	B-cell Maturation Antigen
CAR-T	Chimeric Antigen Receptor T-cell Therapy
CCI	Charlson Comorbidity Index
CI	Confidence Interval
CRS	Cytokine Release Syndrome
FDA	United States Federal Food and Drug Administration
GSK	GlaxoSmithKline
HIPAA	Health Insurance Portability and Accountability Act
ICANS	Immune Effector Cell Associated Neurotoxicity Syndrome
IMiD	Immunomodulatory Agents
LOT	Line of Therapy
MM	Multiple Myeloma
NCCN	National Comprehensive Cancer Network
NDMM	Newly Diagnosed Multiple Myeloma
PI	Proteasome Inhibitors
QC	Quality Control
RRMM	Relapsed Refractory Multiple Myeloma
RWE	Real-World Evidence
RWB	Real-World Biostatistics
SCT	Stem Cell Transplant
SD	Standard Deviation
TE	Transplant Eligible
TI	Transplant Ineligible
TTD	Time to Treatment Discontinuation
TNT	Time to Next Treatment
ULN	Upper Limit of Normal

USUnited States

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1. SYNOPSIS

Title	Ummet Needs and Rates of Key Safety Events Relating to Treatment Use among Newly Treated and Relapsed Refractory Patients with Multiple Myeloma
Background and Rationale	Multiple myeloma (MM) is a progressive clonal plasma-cell malignancy associated with immune dysfunction and recurrent relapse. Within the United States (US) it accounts for roughly 1.7% of newly diagnosed cancers. The five-year survival rate is approximately 64%, with heterogeneous receipt of first-line (1L) and later line treatments often driven by clinical factors such as tolerability concerns. Considering the continued need for effective MM therapies, further foundational work relating to the unmet needs of MM patients associated with treatment receipt is needed.
Study Design and Study Period	This non-interventional study will utilize data from the Optum Clinical Datamart (CDM) database. The study period is defined as 01 Jan 2016 through latest data availability at time of analysis.
Study Objectives	Primary Objectives - To describe the demographics and clinical characteristics of the NDMM (1L) and RRMM (2L and 3L) populations, including background rates of key safety events (i.e., ocular, hepatic, infections) and occurrence of healthcare utilization events among patients prior to the receipt of 1L, 2L, and 3L. - To estimate on-treatment event rates and on-treatment incidence of key safety and healthcare utilization events while on 1L, 2L, 3L, within each LOT and by SOC. Secondary Objectives - To describe treatment patterns and treatment-related outcomes (e.g., attrition, time to next treatment [TTNT], and time to discontinuation [TTD]) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by transplant eligibility status proxies.
Study Population	NDMM and relapsed/refractory patients who have been treated for MM will be assessed according to the receipt of lines of therapy, according to: ● 1L Cohort (i.e., NDMM): newly treated patients in the 1L setting. Receipt of 1L will be utilized as a proxy for NDMM ● 2L Cohort (i.e., RRMM): patients who received 2+ lines of therapy ● 3L Cohort (i.e., RRMM): patients who received 3+ lines of therapy
Outcomes	Primary objectives: (1) baseline demographic and clinical characteristics and baseline prevalence of key safety events and healthcare utilization; (2) incidence of key safety events and healthcare utilization while on 1L-3L regimens: ● Hepatic toxicity ● Renal impairment ● Severe infection ● Cardiovascular events (cardiovascular thromboembolic events) ● Neurological events (movement disorders, neurological conditions) ● Healthcare utilisation (inpatient admissions, emergency visits) Secondary objective: treatment patterns, and treatment related outcomes to serve as a proxy for efficacy and tolerability: ● Time to next treatment

	● Time to discontinuation ● Time to death
Statistical Analyses	Study analyses will be descriptive. Descriptive statistics will be used to summarize baseline data and descriptive outcomes. Kaplan-Meier curves will be used for time-to-event outcomes. No formal hypotheses testing will be conducted.

2. AMENDMENTS AND UPDATES

Amendment or update no	Date	Section of study protocol	Amendment or update	Reason
None	N/A	N/A	N/A	N/A

3. MILESTONES

Milestone	Planned date
Final approval of protocol	10 July 2026
Final approval of table shells	10 July 2026
Interim results for ASH 2027	15 January 2027
Final report of study results	15 July 2027

4. BACKGROUND AND RATIONALE

4.1. Background

Multiple myeloma (MM) is a progressive clonal plasma-cell malignancy characterized by immune dysfunction and recurrent relapse, commonly resulting in end-organ damage including lytic bone disease, anemia, renal impairment and hypercalcemia.¹ MM primarily affects older adults (median age at diagnosis: ~65 years), and remains an important cause of cancer morbidity and mortality worldwide.² In the United States (US), MM represents approximately 1.7% of newly diagnosed cancers, and a similar proportion of cancer deaths.³ Although five-year survival has improved over the past two decades, reaching ~64% in recent 2016-2022 estimates,³ outcomes remain suboptimal because the disease remains largely incurable and survival varies markedly by age, frailty, and transplant eligibility.⁴ Early mortality also remains substantial, particularly among older adults, with approximately 30% of patients dying within one year of diagnosis.⁵

Against this clinical backdrop, the therapeutic landscape for MM has evolved rapidly, but treatment use, response, and outcomes continue to vary across patients. Frontline care differs by transplant eligibility: transplant-eligible (TE) patients frequently receive triplet induction and autologous stem cell transplant (SCT). Many patients, however, are older or have comorbidities that preclude transplant, though heterogeneity in defining transplant eligibility status remains a non-uniform approach.⁶ Among transplant-ineligible (TI) patients, frontline regimens vary and are chosen based on patient characteristics and tolerability, with less efficacious outcomes compared to TE patients.

In RMM, CAR-T and bispecific therapies produce deep responses but are limited by access and tolerability. CAR-T is typically offered to younger, fitter

patients and is concentrated at specialized academic centres, restricting access by patient fitness and geography.⁷ Other later line options, such as bispecifics and BCMA-targeted antibody-drug conjugates (ADCs) like belantamab mafodotin, allow broader outpatient access but have distinct safety considerations. BCMA-targeted ADCs are characterised by ocular adverse events, requiring monitoring and dose modifications.⁸ However, ADCs have a lower incidence of T-cell-mediated toxicities, such as cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), as well as parkinsonian neurotoxicity that characterise CAR-T and some bispecifics.^{9,10}

Compared with more established classes, such as proteasome inhibitors (PI) and immunomodulatory drugs (IMiDs), ADCs have not been characterised primarily by prothrombotic or direct cardiotoxic adverse events.¹¹ However, direct comparative data against bispecific antibodies, PIs, and IMiDs remain limited. Because anti-CD38-based quadruplet regimens often incorporate agents associated with cardiovascular and thrombotic toxicities, comparative evaluation of safety profiles across triplet, quadruplet, and alternative regimens is warranted.

Despite therapeutic advances, unmet needs persist, not only for older and frail patients who have fewer tolerable frontline options, but also in consideration of long-term chronic impacts for younger patients. Trial populations often differ from routine practice, and real-world tolerability and effectiveness can diverge from trial findings. Up-to-date, population-level evidence is therefore needed on background and on-treatment safety rates (for example, ocular, hepatic, and infectious events), healthcare utilization (inpatient and ED visits), treatment uptake and sequencing, time on therapy, and discontinuation. Administrative claims provide a timely, population-level source for these measures.

This non-interventional, retrospective US real-world study will use the Optum Clinical Datamart (CDM) to estimate incidence of key safety and utilization events before and during therapy, describe treatment patterns and sequencing by line of therapy and SOC, and assess treatment outcomes (e.g., TTNT, TTD, and overall survival [OS]) as proxies for effectiveness and tolerability among patients with NDMM and RRMm.

4.2. Rationale

Belantamab mafodotin (BLENREP; GSK2857916) is a BCMA-directed ADC evaluated in the DREAMM clinical programme. The pivotal DREAMM-2 trial demonstrated clinically meaningful single-agent responses in a heavily pretreated RRMm population, and subsequent DREAMM studies (i.e., DREAMM-7 and DREAMM-8)^{12,13} have evaluated belantamab-containing combinations and earlier-line use.

These trial data illustrate both the potential benefits and the practical challenges of BCMA-targeted ADCs, particularly with respect to tolerability and monitoring (for example, ocular toxicity). With continued development (DREAMM-10¹⁴ and DYNAMMIC¹⁵), RWE is needed to further contextualize efficacy and tolerability of alternative non-ADC therapies, and understand unmet need in the treatment landscape.

5. RESEARCH QUESTION AND OBJECTIVE(S)

5.1. Research Question(s)

Investigation of unmet needs with regards to background safety rates according to standard of care (SOC) regimens and treatment utilization and sequencing remains under-characterised. Real-world data can offer insights into safety

rates and treatment patterns, which can be used as a proxy for treatment tolerability and efficacy, demonstrating unmet need.

This study aims to address the following research questions, to be addressed according to 3 cohorts of patients across three nested cohorts defined at each LOT: 1L (proxy for NDMm), 2L (proxy for RRMm), and 3L (RRMm). Each later LOT cohort is a subset of the prior LOT cohort.

- What are the event rates of key safety events (ocular, hepatic, severe infections, inpatient admissions, healthcare utilization) prior to and during each LOT for the NDMm and RRMm settings (i.e., 1L-3L)?
- What are the patterns of treatment receipt, duration, and transition according to standard of care (SOC) regimens in NDMm and RRMm patients, to be assessed by LOT?
- What are patient characteristics associated with unmet need through treatment patterns and outcomes (e.g., older patients, select comorbidities)?

5.2. Objectives

5.2.1. Primary Objective(s)

The primary objectives of this study will assess unmet needs relating to the MM treatment landscape through assessment of treatment tolerability and efficacy as evaluated through rates of key safety events and treatment patterns in the NDMm and RRMm settings. Specifically, the following primary objectives will be addressed:

1. To describe the demographics and clinical characteristics of the NDMm (1L) and RRMm (2L and 3L) populations, including background rates of key safety events (i.e., ocular, hepatic, infections) and occurrence of

healthcare utilization events among patients prior to receipts of 1L, 2L, and 3L

2. To estimate on-treatment event rates and on-treatment incidence of key safety and healthcare utilization events while on 1L, 2L, and 3L, within each LOT and by SOC

5.2.2. Secondary objective(s)

1. To describe treatment patterns and treatment-related outcomes (e.g., treatment attrition, time to next treatment [TTNT], and time to discontinuation [TTD]) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by transplant eligibility status proxies

5.2.3. Exploratory objective(s)

1. To assess use of SOC regimens and SCT over time (i.e., SOC type by year)
2. To assess the treatment patterns through a Sankey diagram from 1L-4L, excluding patients with Medicare
3. To assess the receipt of SCT within 12-months of 1L index (depending on data availability) according to each TI/TE cohort definition
4. To stratify the Primary Objectives and select Secondary Objective results by medical conditions of interest (i.e., cardiovascular history, neurologic history, renal impairment)(to be assessed based on review of data)

6. RESEARCH METHODS

6.1. Study Design

This is a real-world retrospective cohort study using the Optum CDM database. The study will describe patient characteristics, safety rates, and treatment pat-

terns, and will utilize treatment related outcomes (TTNT, TTD, OS) as proxies for tolerability and clinical progression/efficacy to inform areas of unmet need.

The study period, relevant timepoints, and outcomes are described in **Figure 1.**

The study period will span from 01 Jan 2016 through the most recent date of data availability, and all patients will be required to have initiated their first qualifying line of treatment from 01 Jan 2017.

The **index date** will be defined as the start date of a **LOT regimen** (i.e., 1L, 2L, 3L; each have their own index date).

The **baseline period** will be defined as the 12-month period prior to (including) the index date for each LOT. Patient demographics, clinical characteristics, and background safety rates will be summarized during each respective baseline period (i.e., 1L will have a 12-month baseline relative to 1L start, and 2L will have 12-month baseline period relative to 2L start). Only the 1L baseline period will be used to assess eligibility.

The patient **follow-up period** will be defined as the period starting from the index date (i.e., excluding index) until the end of continuous enrollment or end of study period or death, whichever occurs first.

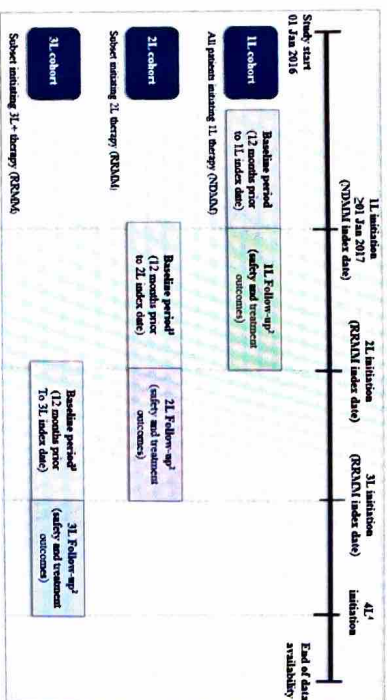
Safety and treatment-related outcomes will be assessed during each LOT follow-up period as described in **Section 6.3.2.**

LOT assignment and regimens will be assigned according to prior internal GSK work to define a claims-based LOT regimen, in brief:¹⁶

- **1L is defined** as all MM therapies received within 60 days of the 1L start date
- **Start of 2L and subsequent LOTs is defined** as the earliest of: a stem cell transplant (SCT; allogeneic or an unplanned autologous SCT), CAR-

T cellular therapy, or the date of the first administration for a new MM agent that was not part of the previous LOT regimen. Each subsequent LOT includes all MM therapies received within 30 days on and following the LOT start date

Figure 1. Summary of Study Design¹



Abbreviations: 1L, first-line; 2L, second-line; 3L, third-line; 4L, fourth-line; NDMM, newly diagnosed MM; RRM, relapsed/refractory MM.

Notes:

- [1] Patients may contribute sequentially to multiple cohorts as they progress through lines of therapy. Three cohorts will be assessed: 1L, 2L, 3L. Each subsequent line is a subset of the prior line. Each cohort contains a 12-month baseline period prior to index date of the LOT, used to assess demographics, clinical, characteristics. Background rates of key safety events and healthcare utilization will also be obtained during each baseline period.
- [2] Key safety events (including healthcare utilization) occurring while on each LOT or within 30 days of discontinuation will be assessed. If a patient discontinued a LOT and experienced an event after 30 days, the event will not be counted for that LOT. Treatment outcomes will be assessed according to the outcome of interest (i.e., TTNT, TTD, OS).
- [3] The baseline periods for 2L and 3L may overlap with time on a prior LOT, depending on the dates of treatment.
- [4] There is no 4L cohort. Only the 4L start date and 4L treatment received will be assessed.

6.2. Study Population and Setting

Newly treated adult patients with MM will be identified in the Optum CDM at the start of 1L. Two nested subset cohorts will be derived from the 1L cohort: patients initiating 2L (e.g., proxy for RRM) and patients initiating 3L (RRM). Full eligibility and operational definitions are provided in Section 6.2.1; subgroup and SOC stratifications are described in Sections 6.2.2 and 6.2.3.

6.2.1. Eligibility Criteria

All adult patients diagnosed with and newly treated for MM following study-specific inclusion and exclusion criteria are eligible for the analysis as members in NDMM cohort (1L cohort). Additional eligibility criteria will be applied to the 1L cohort to create the subset cohorts of patients who received 2+ (2L cohort) and 3+ (3L cohort) lines of therapy.

6.2.1.1. Inclusion Criteria

Overall study eligibility, as defined for 1L NDMM cohort

- **MM diagnosis:** At least one inpatient medical claim with a diagnosis code for MM in any position (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or ≥ 2 outpatient medical claims for MM in any position on the claim, on separate days within 90 days, during the study period
 - As data in Optum CDM is collected primarily for health insurance and not research purposes, all MM diagnosis codes are to be considered. Inclusion will focus on defining NDMM according to patients who are newly treated, defined as the receipt of a 1L treatment
- **Adult age:** Aged ≥18 years in MM diagnosis calendar year

- **Eligible 1L treatment:** Received an eligible 1L treatment (other than belantamab) for MM on or after 01 Jan 2017 (eligible treatment period).
 - The 1L cohort index date is the date of the first claim for MM treatment within the identification period
 - Eligible 1L treatments are defined according to clinical review and include FDA approved regimens, as well as off-label regimens commonly expected for use in 1L. For a full list of eligible 1L therapies, see Appendix 2, with details in **Section 6.2.2.**
- **Continuous enrolment (CE):** CE of at least 12-months with medical and pharmacy benefits before the 1L cohort index date and at least 6 months before the MM diagnosis. Patients with gaps in enrolment of ≤ 30 days are considered to be continuously enrolled

Additional eligibility for 2L and 3L RMM Cohorts

- The subset of subjects with evidence of receiving a second LOT and who are continuously enrolled with medical and pharmacy benefits for at least 12-months before the 2L cohort index date will be included in the 2L cohort at the initiation of 2L
- Patients with evidence of receiving a third LOT and who are continuously enrolled with medical and pharmacy benefits for at least 12-months before the 3L cohort index date will be included in the 3L cohort at the initiation of 3L

6.2.1.2. Exclusion Criteria

- **Evidence of an MM oncology therapy during the 12-month 1L baseline period:** ≥ 1 medical or pharmacy claim for any MM oncology therapy

- This is to ensure that treatment exposure and LOT assignments reflect incident therapy starts at the index date and are not confounded by ongoing or recent prior MM treatments
- **Evidence of another cancer in the 1L baseline period:** Patients with either ≥ 1 inpatient or ≥ 2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer will be excluded (patients with ≥ 2 outpatient claims on separate days within 60 or 90 days apart will also be flagged for consideration)
- **Evidence of pregnancy:** ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the study period
- **Received an ADC (i.e., belantamab) in any LOT**

6.2.2. SOC Categorization and Eligible 1L MM Regimens

To assess treatment patterns and time-to-event treatment outcomes according to LOT cohorts and SOC regimen, regimens will potentially be grouped according to commonly utilized quadruplets, triplets, doublets, anti-CD38 backbone, and class, allowing for potential off-label use.

For purposes of inclusion criteria, eligible 1L treatments will include treatments commonly used or expected in the 1L setting, inclusive of approved, and off-label combination regimens.

Appendix 2 contains an exemplary list of potential treatment combinations that may be included in each group. Final treatment groupings may depend on data availability and distribution of observed treatments and may be recategorized or summarized as 'other', where applicable.

Tentatively, the following SOC groupings are proposed:

• **1L (NDMM):**

- Quadruplets with anti-CD38 backbone
- Triplets with anti-CD38 backbone
- Other triplets (non-anti-CD38)
- Doublets
- Monotherapies

• **Later lines (2L, 3L, 4L):**

- Quadruplets with anti-CD38 backbone (off-label; some use expected)
- Triplets with anti-CD38 backbone
- Other triplets (non-anti-CD38)
- Other novel agents (e.g., Selinexor)
- CAR-T
- Bispecifics (BCMA and non-BCMA)
- Doublets
- Monotherapies (expected to be rare in 2L+)

6.2.3. Stratification by patient subgroups

To assess unmet need according to transplant eligibility, results from the primary and secondary objectives will be stratified by proxies for transplant eligibility among the 1L and 2L patient cohorts only. Patient subgroups will be assessed according to LOT, and will not be assessed according to SOC unless otherwise indicated by the data.

1. Age $\geq$ 70 vs $<$ 70 years

- Age $\geq$ 70 years to be defined as proxy for TI status during baseline, to be compared to age $<$ 70 years as proxy for TE

2. Age and comorbidity burden, defined by Charlson Comorbidity Index (CCI)¹⁷

- Age $\geq$ 70 or CCI $\geq$ 3 to be defined as proxy for TI status during baseline to be compared to Age $<$ 70 and CCI $<$ 3 as proxy for TE

6.3. Variables

All variables including treatment, demographic, and clinical variables will be reported descriptively, and will be defined according to available data within Optum CDM.

6.3.1. Exposure definitions

The exposure of interest is defined as the LOT-based cohort of eligible patients (see **Section 6.2.1** for eligibility criteria). Time-to-event treatment outcomes and safety rates during each LOT will be assessed according to SOC regimens and select patient subgroups.

6.3.2. Outcome definitions

6.3.2.1. Primary outcomes: key safety events and healthcare utilization

Table 1 summarizes all outcome definitions, functional form, and timing of collection for the Primary Objectives. Specifically, Primary Objective 1 will summarize demographics, clinical characteristics and background rates of key safety events during the baseline periods for each 1L, 2L, and 3L cohorts. Primary Objective 2 will assess rates of key safety events while on each LOT (i.e., during the treatment periods).

Key safety events of interest related to MM treatments (i.e., cardiovascular events, neurological events, and renal impairment) will be defined according to

selected ICD-10-CM codes or lab values (**Appendix 3**), applicable lab values, or healthcare visits.

An event will be attributed to a LOT if it occurs between the LOT's start date and the start date of a subsequent LOT, or discontinuation date of the prior LOT + 30 days of discontinuation, whichever comes first. If the patient experiences an event ≥30 days after discontinuation, the event will not be counted (even if the patient eventually started a subsequent LOT).

Key safety outcomes include the following:

• **Hepatologic toxicity will be defined:**

- According to ICD-10-CM diagnosis codes for conditions relating to: Abnormal liver function, Toxic liver disease, Hepatic failure, Chronic hepatitis, Fibrosis and cirrhosis of the liver, Nonalcoholic steatohepatitis, Inflammatory liver disease, and Other diseases of liver, OR;
- According to laboratory values indicative of moderate (serum bilirubin >1.5 x Upper Limit of Normal [ULN] and any aminotransferase [AST]) or severe hepatic impairment (serum bilirubin >3 x ULN and any AST)

• **Renal impairment will be defined:**

- According to ICD-10-CM diagnosis codes relating to renal impairment, chronic or acute kidney disease, or other renal impairment condition, OR;
- According to laboratory values indicative of eGFR<mL/min/1.73 m²

• **Serious infection will be defined as:** infection associated with hospitalization or death within 30 days of infection or where infection is listed as a primary/principal diagnosis for an inpatient record

• **Ocular events will be defined as:** dry eyes, all keratopathies (including ulcerative and infective), blurred vision, ocular pain, cataracts, macular degeneration, glaucoma, retinal vascular occlusion, retinal hemorrhage, diabetic retinopathy, photophobia, eye irritation, corneal ulcer, foreign body sensation)

• **Cardiovascular conditions will be defined as:** heart failure, myocardial infarction, coronary artery disease, arrhythmia, hypertension, venous thromboembolism, stroke/transischemic attack (TIA), cerebrovascular disease, peripheral artery disease

• **Neurologic conditions will be defined as:** peripheral neuropathy, cognitive impairment/dementia, Parkinson's disease or other movement disorders, seizure disorders

Health care utilisation outcomes include the following:

- **Inpatient hospitalization**
- **ER visits**

Table 1. Outcomes, definitions, and timing of collection for primary objectives

Outcomes	Definitions	Timing of Collection
Primary Objective 1: demographics and clinical characteristics of the NDMA (1L) and RRMA (2L and 3L) populations, including background rates of key safety events (i.e., ocular, hepatic, infectious) and occurrence of healthcare utilization events among patients prior to the receipt of 1L, 2L, and 3L.		
Demographic characteristics		
Age	• Continuous (years) • Binary (≥70 vs <70) • Categorical ○ 18-44 years ○ 45-64 ○ 65-74 years ○ ≥75 years	At index calendar year (1L, 2L, 3L)

Outcomes	Definitions	Timing of Collection
	<i>Age categories may be adjusted based on age distribution in the study population</i>	
Sex	Categorical: • Male • Female • Unknown	At index (1L, 2L, 3L)
Region	Categorical: • Midwest • South • West • Northeast • Unknown <i>Based on regions defined by US Census Bureau</i>	At index (1L, 2L, 3L)
Race	Categorical: • Asian • Black • White • Unknown	At index (1L, 2L, 3L)
Ethnicity	Categorical: • Hispanic or Latino • Hispanic or Latino Not Hispanic or Latino • Unknown	At index (1L, 2L, 3L)
Insurance type	Categorical: • Medicare • Commercial Health Plan	At index (1L, 2L, 3L)
<i>Comorbidities and clinical characteristics</i>		
Charlson Comorbidity Index (CCI) ¹⁷	Continuous: Categorical: 0, 1, 2, 3, 4, 5+	During baseline (1L, 2L, 3L)
Year of MM diagnosis	Categorical: number and percent of NDMAM patients by Year of first MM diagnosis First MM diagnosis is defined as first medical claim for MM within the baseline period prior to 1L	Diagnosis date

Outcomes	Definitions	Timing of Collection
Follow-up time from diagnosis	Continuous (months). Time from diagnosis date to patient's follow-up end date	Study period (1L, 2L, 3L)
Follow-up time from index	Continuous (months). Time from index date (included) to patient's follow-up end date	Follow-up period (1L, 2L, 3L)
Year of 1L, 2L and 3L initiation	Categorical: Number and percent of patients by year, from 2017 to 2025 (or latest data availability)	At treatment initiation date (1L, 2L, 3L)
<i>Background rates of key safety events</i>		
Prevalence of key safety events of interest	Categorical: number and percent of patients with occurrence of event of interest (see Section 6.3.2.1 for definitions): • Hepatologic toxicity • Renal impairment • Serious infection • Ocular events • Cardiovascular conditions • Neurologic conditions	During baseline (1L, 2L, 3L)
Background rates of key safety events of interest	Numerical: reported as events per unit person-years (PY). Calculated as the number of events for an outcome during the baseline period (as listed above and in Section 6.3.2.1) divided by the total baseline person-time at risk	During baseline (1L, 2L, 3L)
<i>Background occurrence of healthcare utilization events, as defined according to evidence of a claim</i>		
Proportion of inpatient hospitalizations	Continuous and categorical: 1, 2, 3, 4+ hospitalization(s)	Hospitalizations starting during baseline (1L, 2L, 3L)
Inpatient length of stay (LOS)	Continuous: <i>Inpatient length of stay (LOS), is calculated among those with an in-patient hospitalization from starting date to end date of hospitalization (can extend over several LOT)</i>	Hospitalization starting during baseline (1L, 2L, 3L)
Proportion of ER visits	Continuous and categorical: 1, 2, 3, 4+ visit(s)	During baseline (1L, 2L, 3L)
Primary Objective 2: on-treatment event and incidence rates of key safety and healthcare utilization events while on 1L, 2L, and 3L, within each LOT and by SOC		

Outcomes	Definitions	Timing of Collection
On-treatment event rates of safety events	Same as noted for baseline analysis Counting events within the treatment period. An event will be attributed to a LOT if it occurs between the LOT's start date and the start date of a subsequent LOT, or discontinuation* date of the prior LOT + 30 days of discontinuation, whichever comes first: • For patients who received a subsequent LOT: events occurring on the first LOT index up to the day before the subsequent LOT will be counted • For patients who discontinued the index LOT prior to a subsequent LOT: events occurring with 30 days of discontinuation will be attributed to the index LOT • Any events occurring after 30 days of discontinuation, even if the patient started a subsequent LOT, will not be considered *per GSK LoT algorithm definition, discontinuation of a regimen occurs when all MM agents in the LoT are stopped	During LOT treatment period (1L, 2L, 3L)
Incidence of key safety events	Numerical, reported as events per units PYs Calculated as the number of incident events for a given outcome divided by the total number of patients at risk for an event (i.e., calculated among patients who did not have a baseline history in the 12-month baseline period prior)	During LOT treatment period (1L, 2L, 3L)
Healthcare utilization events	Same as Primary Objective 1	During LOT treatment period (1L, 2L, 3L)

6.3.2.2. Secondary objective: treatment patterns and treatment-related outcomes

The secondary objective to describe treatment patterns and treatment related outcomes will be addressed through descriptive statistics and time-to-event analyses. Thus, unmet need relating to poor treatment outcomes will be proxied through treatment switching (TTNT) discontinuation (TTD), OS, and attrition.

□

To further understand unmet need relating to patient characteristics, treatment outcomes will be assessed by patient characteristics (i.e., subgroups).

Table 2 summarizes all outcome definitions, functional form, and timing of collection for the Secondary Objective.

Table 2. Outcomes, definitions, and timing of collection for Secondary Objective

Outcomes	Definitions	Timing of Collection
Secondary Objective 1: describe treatment patterns and treatment-related outcomes (e.g., treatment attrition, TTNT, TTD, OS) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by transplant eligibility status proxies		
Treatment patterns		
Patients receiving each line	Number and percent of patients receiving each 1L, 2L, 3L, and 4L regimens	-
Treatment regimens received	Categorical: regimen categories (Section 6.2.2)	Described for overall sequence from 1L to 4L
Switch between successive LOTs	Sankey diagram of switch between regimen categories	Follow-up period
Treatment attrition	Number and percent of patients who received each subsequent LOT, discontinued treatment and did not receive another, were lost to follow-up, or died	Described for overall sequence from 1L up to 4L start
Time from diagnosis to 1L initiation	Continuous (months): Time from diagnosis date (included) until index date (included)	1L index
Time from prior LOT to next LOT initiation	Continuous (months): Defined among patients initiating a subsequent LOT as time from prior LOT start date (included) to next LOT start date (excluded)	• 1L start to 2L start • 2L start to 3L start • 3L start to 4L start
Treatment-related outcomes		

Time to next treatment (TTNI)	Time-to-event outcome, Time from index LOT start date to the earliest between the start of the next LOT or death. Patients without a subsequent LOT or date of death will be censored at their follow-up end date	During each LOT (1L, 2L, 3L)
Time to treatment discontinuation (TTD)	Time-to-event outcome, Time from index LOT start date (included) to the date of treatment discontinuation (included) The discontinuation date is the earliest of the date of treatment discontinuation (end of current LOT), initiation of the next LOT, or death. Patients without treatment discontinuation, next LOT or death will be censored at their follow-up end date	During each LOT (1L, 2L, 3L)
Time to death	Time-to-event outcome, Time from LOT start date to date of death. Patients without a recorded date of death will be censored their follow-up end date	Follow-up period

6.3.2.3. Exploratory objectives: outcomes, definitions, and timing

This study will assess several exploratory outcomes, described below in Table

3.

Table 3. Outcomes, definitions, and timing of collection for Exploratory Objectives

Outcomes	Definitions	Timing of Collection
Exploratory Objective 1: assess trends in the use of SCT over time, overall and according to SOC of SCT		
Trends in the use of SCT	Number of patients with an SCT in 1L and 2L by year, according to SOC type	Follow-up period
Exploratory Objective 2: assess the receipt of SCT within 12-months of 1L index (depending on data availability) according to each T1/T2 cohort definition		
SCT receipt	Number of patients who got at least one SCT within 12-months of 1L index	Follow-up period
Exploratory Objective 3: assess treatment patients through a Sankey diagram from 1L-4L, excluding patients with Medicare		
Treatment switches across LOT for	Sankey among patients on commercial health plans at 1L	Follow-up period from 1L to 4L

commercially insured patients

Exploratory Objective 4: assess unmet need according to baseline medical conditions of interest (i.e., cardiovascular history, neurological history, renal impairment)(to be assessed based on review of data, where applicable)

Stratify Primary Objectives 1 and 2	Stratify the main analyses for Primary Objectives 1 and 2, according to 1L and 2L by pre-existing baseline history of cardiovascular, neurological, and renal impairment conditions	Defined according to each outcome
Stratify Secondary Objective 1 - Treatment Outcomes only	Stratify treatment outcomes (not including Sankey), for the Secondary Objective 1 according to according to 1L and 2L by pre-existing baseline history of cardiovascular, neurological, and renal impairment conditions	Defined according to each outcome

6.3.3. Confounders and effect modifiers

N/A: This analysis is descriptive only.

6.4. Data sources

The Optum CDM database will be utilized for this study. The Optum database is a statistically deidentified, closed system of administrative health claims submitted for payment by providers and pharmacies for US-based commercial health plans and Medicare Advantage members. The database includes the following: (1) patient enrollment; (2) physician, facility, and pharmacy claims. All patients in this database have both medical and pharmacy coverage, allowing analysis of overall healthcare utilization.

Information is available for patients across all care settings including, for example, inpatient visits, emergency department visits, physician office visits (both primary care physician and specialty visits), and prescription medications filled. Available data covers approximately 15-20 million covered lives from 2007 to the present date (estimate current as of 2023), and contains representation from all 50 states. All data is compliant with the Health Insurance Portability and Accountability Act of 1996 (HIPAA) 18

Information for diagnoses of interest, including MM will be collected through claims-based diagnosis tables, while treatment data will derive from pharmacy claims. The pharmacy claims table contains claims submitted by pharmacies for prescriptions filled on an outpatient basis, and includes information on the generic and brand classifications, quantity, and fill dates.

6.5. Study size

Since the main objective of the study is descriptive, it is expected the predicted sample size will be sufficient to produce meaningful results with an appropriate estimate precision. We expect that the Optum CDM will have approximately XXX patients receiving 1L therapy, with approximately XXX patients receiving 2L therapy, and XXX receiving 3L therapy.

Table 4 presents the levels of precision achieved (i.e., width of estimated 95% CI) with those numbers of patients, for proportion estimates of categorical variables in describing patients' clinical and demographic characteristics of the primary objective. The precision levels were computed using PASS 2019 based on the two-sided 95% Clopper–Pearson exact confidence interval for a single binomial proportion.¹⁹

For example, for the 3L cohort (smallest number of patients) with size estimated to be = xxx, if estimated proportion is 10%, this proportion would be estimated with a 95% CI of (xx% to xx%) (i.e. xx width), which is considered as acceptable for this study.

Table 4. Precision estimates for proportions

Sample size	Proportion (p)	Precision (Width of 95% CI)
XXX (1L)	p = 0.1	
XXX (2L)	p = 0.1	
XXX (3L)	p = 0.1	

XXX (1L)	p = 0.2	
XXX (2L)	p = 0.2	
XXX (3L)	p = 0.2	
XXX (1L)	p = 0.3	
XXX (2L)	p = 0.3	
XXX (3L)	p = 0.3	
XXX (1L)	p = 0.4	
XXX (2L)	p = 0.4	
XXX (3L)	p = 0.4	
XXX (1L)	p = 0.5	
XXX (2L)	p = 0.5	
XXX (3L)	p = 0.5	

For stratifications (see SOC and patients subgroups stratification in Sections 6.2.2 and 6.2.3), we will consider not generating analysis when there are <25 patients as the results may not be meaningful.

6.6. Data management

All data analysis will be conducted using R version 4.5.2.²⁰ To derive variables required for analysis, data from Optum CDM will be subset to the population of interest. Information relating to potential eligible patients according to LOT will derive from the previously defined LOT algorithm, developed in-house the real-world biostatistics (RWB) team at GSK.¹⁶

The GSK RWB team will create the analytical dataset and derive variables from the structured data within Optum CDM required to meet the study objectives. ICD-10-CM codes used to identify eligible diagnoses, and NDC codes to identify treatments will be extracted to define indexes, cohorts, and outcomes.

Where applicable, lab values and healthcare utilization data may also be used to defined outcomes.

Data extraction, transformation, and preparation for analysis will be performed using standardized and validated workflows. This includes:

- Data cleaning and validation
- Consistency checks
- Derivation of study variables (e.g., CL, cohort definitions)
- Preparation of analysis dataset
- Quality control

All study data, derived datasets, and analysis outputs will be securely stored and archived in accordance with applicable data protection regulations and internal quality standards. The database is maintained within a secure IT environment, with controlled access, audit trails, and data protection measures in place to ensure confidentiality and integrity.

6.7. Data analysis

All outcomes will be assessed according to their stated objectives. For continuous variables, the descriptive statistics will include means, standard deviations, medians, interquartile ranges, and minimum and maximum values (as applicable). For categorical variables, frequencies and percentages will be generated. Rates will be described in units of PYs, defined as per 10,000 or 100,000 (or other multiplier), depending on data availability.

The number of patients with unknown/missing values for continuous and categorical variables will be reported. No statistical comparisons or p-values will be reported.

If there are less than 25 patients in a particular stratifications or cohort, analyses will not be conducted.

6.7.1. Primary analyses

Primary objective 1: demographics and clinical characteristics of the NDMM (1L) and RMM (2L and 3L) populations, including background rates of key safety events (i.e., ocular, hepatic, infections) and occurrence of healthcare utilization events among patients prior to the receipts of 1L, 2L, and 3L

Patient demographic and clinical characteristics

Baseline demographic and clinical characteristics will be summarized according to the 1L, 2L, and 3L cohorts. These results will be further assessed according to SOC.

All baseline characteristics, except for MM diagnosis date, and comorbidities will be assessed at the time of index date where possible. If data is missing at index, data from the baseline period present nearest index will be used. Comorbidities will be assessed over the 12-month baseline period, including the index date.

Additional subgroup stratifications, as defined in the exploratory analyses may be applied, depending on review of data and determination of study needs.

Background rates of key safety events

Key background prevalence rates of the pre-existing conditions described in **Section 6.3.2.1** will be computed and tabulated for cohorts 1L-3L overall and stratified by SOC.

Background rates of events and corresponding 95% confidence intervals (CI), will be calculated according to the following, where X is the event type/safety outcome of interest (cardiovascular, neurological, renal impairment, etc.), and Y is the selected index LOT (1L, 2L, 3L).

Event rate for event type X =
$$\frac{\text{No.of Event Type X Occuring LOT Y Baseline Period}}{\text{Total PY at Risk During Baseline Period}}$$

Background rates health care utilization events

The number and proportion of patients with at least one hospitalization or an ER visit during the baseline period will be computed and tabulated for cohorts 1L-3L overall and stratified by SOC.

Length of stay will be computed from start of baseline until end of hospitalization. Hospitalization starting before the start date of baseline or ending after the date of baseline will be counted on the overall visit level, and will qualify for their respective time period.

Primary Objective 2: rates of the same key safety events while on 1L, 2L, and 3L will be computed for cohorts 1L-3L overall and stratified by SOC

Event rates and incidence of safety events

Event rates and 95% CI will be similarly calculated among the overall population in each LOT, according to:

Event rate for event type X =
$$\frac{\text{No.of Event Type X Occuring LOT Y Treatment Period}}{\text{Total PY at Risk Among Cohort Sample During Baseline Period}}$$

Prevalence rates will also be computed on patients who have ≥3 months of follow-up from their index date to ensure adequate time in the database for safety and health care utilization assessments.

Additionally, incidence rates will be computed among only participants within a prior history of a prior event at the time of the index LOT according to:

Incidence of event type X =
$$\frac{\text{No.of Event Type X Occuring LOT Y Treatment Period}}{\text{Total PY at risk among those with no prior history at time of LOT}}$$

To compute the number of hospitalizations and length of stay, only the hospitalizations starting in each of the considered LoT will be selected.

6.7.2. Secondary analyses

Secondary Objective 1: describe treatment patterns and treatment-related outcomes (e.g., treatment attrition, time to next treatment [TTNT], and time to discontinuation [TTD]) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by select subgroups

Treatment patterns and regimens received

The proportion of each 1L through 4L regimens received will be described descriptively, as the count and percentage of each regimen.

Treatment sequencing in the Overall Cohort will be presented using Sankey diagrams, with corresponding tables showing frequency and proportion of treatment type. The Sankey diagram will illustrate transitions between SOC treatments from one LOT to the next, or to no subsequent treatment due to death or censoring at follow-up end date.

Overall patient attrition (e.g., number of patients who received only 1L, died, or switched to 2L) will be depicted and tabulated. These analyses may be stratified by patient subgroups, according to SOC.

Time-to-event treatment related outcomes

Outcomes will only be assessed in the subset of patients who have ≥3 months of potential follow-up from their index date to ensure adequate time in the database for outcome assessments.

Treatment related time-to-event analyses (e.g., TTNT, TTD, OS) will be performed using the Kaplan-Meier (KM) product limit estimator. Median survival estimates and Brookmeyer-Crowley 95% CI will be the primary estimate reported.²¹ Additionally, the number and percentage of patients at risk, with an event and censored will be reported along with survival probabilities (with 95% CI) at relevant landmarks, such as 6, 9, 12, 18, and 24 months after index date. Time zero, or the index date, will be 2L start for all TTE outcomes in the primary analyses. Results will be reported in a tabular format along with corresponding KM curves. No log-rank or hypotheses testing will be performed to assess for differences across strata.

6.7.3. Exploratory analyses

6.7.4. Data handling conventions

All study outcomes relating to diagnoses will be defined according to pre-defined code lists, as specified in the **Appendix 3**. Observations where data is missing will be dropped when necessary. No imputation for missing data will be performed.

6.8. Quality control and Quality Assurance

This study will be conducted according to local quality control processes. This procedure requires documented evidence that the study protocol has been correctly interpreted and executed.

This study will follow established quality control and quality assurance procedures, including the following:

Analytical quality control

- Review of data extraction and transformation processes
- Independent verification of statistical code and outputs
- Reproducibility checks

Data storage and security

- Secure IT environment with controlled access
- Audit trails and data protection measures
- Archiving of datasets and analysis outputs

6.9. Limitations of the research methods

This study should be interpreted in light of its limitations, including limitations inherent to the use of claims-based data. First, this study utilized data available in Optum CDM. Optum primarily offers data from a single private payer, with a small proportion of Medicare advantage plans, thus data is limited to members within these plans and does not represent practices or care receipt among other insurance types or plans. However, it is not expected that this restriction would result in different clinical care of patients and receipt of treatments.

LOTs are not explicitly captured in claims data and the algorithms used to identify lines of therapy were based on pre-defined assumptions. While careful consideration, including alignment to treatment guidelines and expert review was applied to define the algorithm, it is possible that decision rules may not directly align to a patients' medical record or actual care. Regardless, claims-based indicates treatments submitted for reimbursement, which is commonly used as a proxy for receipt of care.

As with retrospective research, the variables and outcomes selected are limited to the underlying data structure and availability. As a result, TTNT and TTD

were used as a proxy for efficacy and tolerability of treatment regimens and further clinical outcomes such as PFS could not be ascertained. As a result, possible explanatory data, such as reasons for discontinuation or granular adverse event information are not included, but could alternatively explain results. For example, it is possible that a patient who only received 1L and stopped had a positive outcome or went into remission, while the current study would interpret this finding as potential poor efficacy. However, due to the nature of MM, it is expected that attrition concerns would likely be negative.

7. PROTECTION OF HUMAN SUBJECTS

7.1. Ethical approval and subject consent

This study will comply with all applicable laws regarding participant privacy. No direct subject contact or primary collection of individual human subject data will occur. Study results will be in tabular form and aggregate analyses that omits subject identification, therefore informed consent, ethics committee or IRB approval are not required. Any publications and reports will not include subject identifiers.

7.2. Subject confidentiality

The confidentiality of records that could identify patients within the database must be protected, respecting the privacy and confidentiality rules in accordance with the applicable regulatory requirement(s).

8. LEGAL BASIS FOR PROCESSING INDIVIDUAL HUMAN DATA

The authors confirm that study data is Individual Human Data (IHD) not owned by GSK/ViiV, but that the proposed use of the IHD aligns with the 'purpose of

use' outlined in the source contract and/or the terms and conditions of use of the data source and it will comply with any specified prohibitions of use.

9. MANAGEMENT AND REPORTING OF ADVERSE EVENTS/ADVERSE REACTIONS

Study type 4: Secondary data collection studies based on structured data or unstructured data without human review without safety objectives. Structured data includes de-identified/anonymised healthcare data or health data are converted to a structured forms using automated/computer algorithms e.g., natural language processing.	<i>These studies will not identify safety events as per study objective and cannot identify spontaneous events</i>
Unstructured data without human review, includes use of an automated algorithm and/or natural language processing for data extraction	

There is no potential to collect serious and non-serious AEs, pregnancy exposures, or incidents related to any GSK/ViiV product during the conduct of this research, as the minimum criteria of identifiable patient, reporter, exposure and event, needed to collect and report individual case safety reports are not present in the data source. The study is based on secondary anonymised healthcare data which lack an identifiable patient and reporter and are insufficient to establish attribution between a potential safety event and an individual patient using a GSK/ViiV product.

10. PLANS FOR DISSEMINATING AND COMMUNICATING STUDY RESULTS

Authorship of study manuscripts and presentations at scientific conferences will follow the guidelines established by the International Committee of Med-

ical Journal Editors (<https://www.icmj.org/>). It is intended that one or more peer-reviewed publications will be developed.

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ANNEX 1 LIST OF STAND-ALONE DOCUMENTS

No.	Description	Document Reference No	Date
1.	List of stand-alone documents		
2.	Categorization of SOC regimens		
3.	Codists to define study outcomes		<<Date>>
4.	Main study table shells	<<No>>	<<Date>>
5.	Main study figures		
6.	Algorithm to define LoTs	<<No>>	<<Date>>

ANNEX 2 CATEGORIZATION OF SOC REGIMENS

ANNEX 3 TABLES

ANNEX 4 FIGURES

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As with table shells, a list of figures and shells should be provided to make the anticipated study results clear and transparent.

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ANNEX 5 CODELISTS

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ANNEX 6 LOT ALGORITHM

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